MILITARY SPECIFICATION

ANTENNA COUPLER GROUP, AN/SRA-56, AN/SRA-57, AND AN/SRA-58

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, performance, and testing of a 1 kilowatt (kW) (average), 2 kW peak envelope power (PEP) high frequency (HF) transmit Antenna Coupler Group, AN/SRA-56 (2 megahertz (MHz) to 6 MHz), AN/SRA-57 (4 MHz to 12 MHz), and AN/SRA-58 (10 MHz to 30 MHz), hereinafter referred to as the antenna coupler group.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment And Systems, Requirements For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-W-16878	Wire, Electrical, Insulated, General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts; Packaging And Packing Of
MIL-F-19207	Fuseholder, Extractor Post Type, Blown Fuse Indicating And Nonindicating, General Specification For
MIL-E-21981	Electronics Type Designations, Identification Plates And Markings; Requirements For
MIL-C-24056	Cabinets, Electronic Equipment, CY-4516(A)/S And CY-4516(B)/S (Single Bay) (Naval Shipboard Use)
MIL-C-39012	Connectors, Coaxial, Radio Frequency, General Specification For
MIL-C-39012/6	Connectors, Plug, Electrical, Coaxial, Radio Frequency, (Series C (Cabled), Male, Class 2)
MIL-C-39012/7	Connectors, Plug, Electrical, Coaxial Radio Frequency, (Series C, (Cabled), Female, Class 2)
MIL-C-39012/8	Connectors, Receptacle, Electrical, Coaxial, Radio Frequency, (Series C, (Cabled), Female, Flange Mounted, Rear Mounted, Class 2)
MIL-C-39012/9	Connectors, Coaxial Radio Frequency Series C (Cabled-Receptacle, Female, Jam Nut, Front Mounted, Class 2)
MIL-C-39012/10	Connectors, Plug, Electrical, Coaxial, Radio Frequency, (Series C, (Cabled), Male, Right Angle, Class 2)
MIL-C-39012/11	Connectors, Receptacle, Electrical, Coaxial, Radio Frequency, (Series C, (Cabled), Female, Jam Nut, Rear Mounted, Pressurized, Class 2)
MIL-C-39012/12	Connectors, Coaxial, Radio Frequency, Series C (Uncabled-Receptacle, Female, Flange Mounted, Rear Mounted, Class 2)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-8111), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-C-39012/13	Connectors, Coaxial, Radio Frequency, Series C (Uncabled-Receptacle, Female, Jam Nut Mounted, Front Mounted, Pressurized, Class 2)
MIL-C-39012/14	Connectors, Coaxial, Radio Frequency, Series C (Uncabled-Receptacle, Female, Jam Nut Mounted, Hermetic Seal, Class 2)
MIL-C-39012/15	Connectors, Plug, Electrical, Coaxial, Radio Frequency, (Series C (Cabled), Male, Class 2)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-965	Parts Control Program
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)

HANDBOOKS

MILITARY

MIL-HDBK-235	Electromagnetic Radiated Environment Considerations For Design and Procurement Of Electrical And Electronic Equipment, Subsystems And Systems
MIL-HDBK-237	Electromagnetic Compatibility Management Guide For Platforms, Systems And Equipment
MIL-HDBK-241	Design Guide For Electromagnetic Interference (EMI) Reduction In Power Supplies

(Copies of specifications, standards, and handbooks required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The antenna coupler group shall be in accordance with MIL-E-16400, unless otherwise specified herein. The antenna coupler group shall be designed for transmit operations in the 2 MHz to 30 MHz frequency range. The AN/SRA-56 shall be designed to connect up to four transmitters to one antenna or a maximum of eight transmitters when the associated impedance matching network is used. The AN/SRA-57 shall be designed to connect up to four transmitters to one antenna or a maximum of eight transmitters when the associated impedance matching network is used. The AN/SRA-58 shall be designed to connect up to four transmitters to one antenna.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 Operational requirements. Each antenna coupler group shall give reliable performance under conditions of intermittent or continuous operation during the exposure to adverse conditions normally encountered in Naval ship communication service. Mechanical and electrical ruggedness is of paramount importance for assurance of reliable performance without failure.

3.2 Characteristics. Antenna coupler group characteristics shall be as specified in 3.2.1 through 3.2.3.5.

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3.2.1 Performance characteristics. Performance characteristics shall be as specified in 3.2.1.1 through 3.2.1.7.

3.2.1.1 Input parameters. The antenna coupler groups shall be capable of handling 4 kW (average), 8 kW (PEP). The impedance for each multicoupler unit shall be nominally 50 ohms, unbalanced.

3.2.1.2 Operating frequency range. The antenna coupler group shall be capable of operating over the frequency range 2 MHz to 30 MHz. Each antenna coupler group shall include couplers capable of being tuned over the frequency ranges specified in TABLE I.

TABLE I. Antenna coupler frequency ranges (MHz).

Antenna coupler	Frequency low limit	Range nominal	Frequency high limit
AN/SRA-56	1.96	2 to 6	6.15
AN/SRA-57	3.90	4 to 12	12.30
AN/SRA-58	9.75	10 to 30	30.75

3.2.1.3 Passband characteristics. Passband characteristics shall be as specified in 3.2.1.3.1 and 3.2.1.3.2.

3.2.1.3.1 Insertion loss. Insertion loss shall be less than 2 decibels (dB) between the input connector and the antenna output connector (four or eight channels) in a 50-ohm system.

3.2.1.3.2 Bandwidth. The bandwidth of each coupler unit shall be equal to or greater than 12 kilohertz (kHz) at the 3 dB points.

3.2.1.4 Isolation characteristics. Isolation characteristics shall be as specified in a through c:

a. The isolation between the input terminals of any two operating channels (four channels or eight channels) shall be at least 35 dB for frequency separation equal to, or greater than, 5 percent throughout the frequency range of the multicoupler.

b. The isolation between the input terminal of any operating channel and the output terminal of the multicoupler (four channels or eight channels) shall be at least 35 dB for frequency separations equal to, or greater than, 5 percent of the operating frequency throughout the frequency range of the multicoupler.

c. The isolation between the input terminal of any operating channel and the output terminal of the multicoupler (four channels or eight channels) shall be at least 60 dB at all frequencies between 2 MHz and 30 MHz that are equal to, or greater than, 25 percent of the operating frequency.

3.2.1.5 Output impedance. Each coupler shall provide an output impedance of 50 ohms, unbalanced.

3.2.1.6 Input impedance. Each coupler shall provide an input impedance of 50 ohms, unbalanced. The voltage standing wave ratio (VSWR) input for each individual channel shall be less than 1.05:1.0 for any 3:1 VSWR load.

3.2.1.7 Radio frequency (RF) overload protection. Each coupler unit shall include overload protection and indication to:

a. Limit RF voltage in the output resonator to 6.25 kilovolts (kV) root-mean-square (rms) and the reflected power in the transmitter to a 3:1 ratio at a 1-kW forward power level (250 watts) reflected power.

b. Provide visual local and remote indication of overload protection conditions. The overload protection and indication circuits shall be actuated automatically when an RF signal is transmitted within ±3 percent of the tuned frequency.

3.2.2 Physical characteristics. Physical characteristics of the antenna coupler group shall be as specified in 3.2.2.1.

3.2.2.1 Dimensions and weight. Dimensions and weight shall be as specified in TABLE II.

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TABLE II. Dimensions and weight.

Items	Width (in.) 1/	Depth (in.)	Height (in.)	Maximum nominal weight (lbs) 2/
Antenna Coupler Drawer Units, CU-1772/SRA-56, CU-1774/SRA-57, CU-1776/SRA-58	19.0625	30.125	14.00	88
Antenna Coupler Combiner Units, CU-1773/SRA-56, CU-1775/SRA-57	32.00	8.50	15.125	30
Antenna Coupler Group, AN/SRA-56, AN/SRA-57, AN/SRA-58	23.00	38.50	77.00	735 3/
Dummy load	N/A	N/A	N/A	20 4/

1/ Inches
2/ Pounds
3/ Exclusive of dummy load, packing, crating, maintenance, and installation kits
4/ Exclusive of packing, crating, maintenance, and installation kits

3.2.3 Environmental service conditions. Environmental service conditions of the antenna coupler group shall be as specified in 3.2.3.1 through 3.2.3.5.

3.2.3.1 Operating temperature range. The antenna shall have the operating temperature range specified in MIL-E-16400 for the equipment used in a sheltered controlled environment (Range 4).

3.2.3.2 Nonoperating temperature. The antenna shall have the nonoperating temperature range specified in MIL-E-16400 for the equipment used in a sheltered controlled environment (Range 4).

3.2.3.3 Shock. The antenna coupler shall be capable of withstanding the shock requirements of MIL-S-901 for Grade A, Class III, medium weight, Type A for shipboard equipment.

3.2.3.4 Vibration. The antenna coupler group shall withstand the Type I vibration test of MIL-STD-167-1 at an upper vibration frequency limit of 50 hertz (Hz). Vibration test frequencies shall be selected randomly, not in incremental steps.

3.2.3.5 Salt fog. The antenna coupler group shall withstand the salt fog test for sheltered equipment of MIL-E-16400.

3.3 Parts and materials. Parts and materials shall be as specified in 3.3.1 through 3.3.2.

3.3.1 Parts. Parts shall be as specified in 3.3.1.1 through 3.3.1.4.

3.3.1.1 Parts controls. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.3.1.2 Tuning capacitors. All tuning capacitors shall have an air dielectric and shall conform to MIL-STD-454, Requirement 2. Tuning capacitors shall have midline or straight line frequency tuning characteristics. Alignment provisions shall be incorporated for each tuning capacitor.

3.3.1.3 RF input and output (I/O) connectors. The coaxial RF connectors for the receiving antenna inputs and receiver antenna output shall be Series C in accordance with MIL-C-39012 and MIL-C-39012/6 through MIL-C-39012/15. The receptacles shall be mounted at the top of the equipment cabinet and shall be suitably labeled. The center pin of the Series C connectors shall be made captive in all directions. RF inserts, if used, shall be constrained from nonperpendicular movement to ensure proper mating. Connectors shall be of the solder-lug type; crimped pin types shall not be used.

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3.3.1.4 Multiconductor connectors. Mating plugs for multiconductor cable connectors shall be provided with cable clamps. All external mounted connector receptacles shall be provided with a suitable protective plastic cap to prevent damage to the connector when the mating connector is not installed. Separate internal and external connectors shall be used for the application specified in a through c:

a. Primary alternating current line voltage
b. Receiver audio input
c. Remote overload indicator

3.3.2 Materials. No ferrous material or nonlinear part shall be used in the main RF signal path. Nonlinear parts shall not be connected to the input resonator for use with overload detection or protection.

3.4 Electrical design and construction. Electrical design and construction of the antenna coupler group shall be as specified in 3.4.1 through 3.4.2.

3.4.1 Electrical power characteristics. The antenna coupler group shall conform to the interface requirements of DoD-STD-1399, Section 300. Nominal supply voltage shall be 115 volts alternating current (VAC), 60 Hz, single-phase, Type I.

3.4.1.1 Steady-state voltage and frequency. The antenna coupler group shall conform to the maximum dissipation requirements of MIL-E-16400 under the conditions specified in a through f:

a. Nominal voltage: 115 VAC
b. Lower voltage limit: 103.5 VAC
c. Upper voltage limit: 126.5 VAC
d. Nominal frequency: 60 Hz
e. Lower frequency limit: 57 Hz
f. Upper frequency limit: 63 Hz

3.4.2 Internal wiring and cabling. All wiring shall be polytetrafluoroethylene insulated. Hookup wire shall conform to MIL-W-16878, Type E.

3.5 Mechanical design and construction. Mechanical design and construction shall be as specified in 3.5.1 through 3.5.5.

3.5.1 Rounded corners and edges. All edges and corners of the enclosure shall be rounded. Sharp edges and points of any kind shall be avoided.

3.5.2 Equipment mounting. The antenna coupler group shall be deck-mounted in accordance with MIL-E-16400. Resilient mounts shall not be used.

3.5.3 Composition. The antenna coupler group shall consist of four individual coupler units, a power supply unit, and an impedance matching network. The coupler units and power supply unit shall be slide-mounted in a cabinet. The slides shall be equipped with automatically operated locks which lock units in both servicing and operating positions. Design similarities are specified in TABLE III.

TABLE III. Basic similarities in antenna coupler groups.

Items	Antenna coupler groups		
	AN/SRA-56	AN/SRA-57	AN/SRA-58
Antenna coupler units	CU-1772/SRA-56	CU-1774/SRA-57	CU-1776/SRA-58
Power supply	PP-4993/SRA	PP-4993/SRA	PP-4993/SRA
Equipment cabinet	CY-6422/SRA-56	CY-6423/SRA-57	CY-6424/SRA-58
Impedance matching network	CU-1773/SRA-56	CY-1775/SRA-57	NONE
Frequency range (MHz)	1.95 to 6.15 2 to 6 (nominal)	3.90 to 12.30 4 to 12 (nominal)	9.75 to 30.75 10 to 30 (nominal)

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TABLE III. Basic similarities in antenna coupler groups. (continued)

Items	Antenna coupler groups		
	AN/SRA-56	AN/SRA-57	AN/SRA-58
Channels per group maximum configuration	4 8	4 8	4 4
Maximum RF power per channel	1 kW average 2 kW peak	1 kW average 2 kW peak	1 kW average 2 kW peak
Primary power	115 VAC	115 VAC	115 VAC

3.5.3.1 Cabinet. The electrical cabinet equipment shall conform to the requirements of MIL-C-24056. The cabinet is the basic antenna coupler group enclosure, but also provides the RF and electrical power interface requirements to the coupler units and the power supply unit.

3.5.3.2 Coupler units. The coupler units are basically composed of a front panel (containing tuning control, frequency dial, overload indicator), a resonator assembly, an RF interface, and a power connector assembly.

3.5.3.3 Dummy loads. Each four-channel multicoupler shall be provided with a 1 kW rms (continuous duty) 50-ohm internal dummy load for tuning. THe VSWR shall not exceed 1.2 over the frequency range.

3.5.4 Controls, indicators, and panel layout. The controls, indicators, and panel layout shall be as specified in 3.5.4.1 through 3.5.4.2.5.

3.5.4.1 Coupler unit control. Each coupler unit shall be equipped with the controls specified in a through e:

a. Tune and operate switch
b. Frequency adjust input (tuning control)
c. Frequency adjust output (tuning control)
d. Output coupling
e. Emergency switch

3.5.4.1.1 Tune and operate switch. The tune and operate switch shall be a five-position rotary switch that controls the sequence of tuning and operation, the selection of control and monitoring devices, and the application of protective elements as required to tune and operate the antenna coupler. The designated positions shall be as specified in a through e:

a. OFF
b. Tune-1
c. Tune-2
d. Tune-3
e. Operate

3.5.4.1.2 Frequency adjust I/O control. The frequency adjust I/O controls shall be rotary controls that adjust the resonant frequency of the I/O resonator. Decrease or increase panel markings indicate the direction that the frequency adjust I/O controls shall be rotated to decrease or increase the resonant frequency. The lock or unlock dial permits the controls to be secured at any desired frequency. The tuning of the resonator over the 3:1 frequency range shall require a minimum of 60 turns of the tuning controls and there shall be not more than 3 in.-lb of torque required to turn the controls.

3.5.4.1.3 Output coupling control. The output coupling control shall be a rotary control that adjusts the inductive coupling between the output resonator and output link. Decrease or increase panel markings indicate the direction that the output coupling control must be rotated to decrease or increase the output coupling. Lock or unlock dial permits the control to be secured in any desired position. The adjustment of the output coupling from minimum to maximum shall be a minimum of 20 turns of the coupling control.

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3.5.4.1.4 Emergency ON and OFF switch. The emergency ON and OFF switch shall be a two-position toggle switch that bypasses protective circuits activated during a cooling system failure, overvoltage, or excessive VSWR to permit the antenna coupler to operate in an emergency mode.

3.5.4.1.5 Frequency adjust I/O dials. The frequency adjust I/O dials shall be horizontally moving, flexible steel, tape dials calibrated from 1.9 MHz to 6.4 MHz (AN/SRA-56), 3.9 MHz to 13.0 MHz (AN/SRA-57), or 9.5 MHz to 32.0 MHz (AN/SRA-58) that indicate the frequency to which the I/O resonators are tuned.

3.5.4.2 Indicators. Indicators shall be as specified in 3.5.4.2.1 through 3.5.4.2.5.

3.5.4.2.1 Tuning dial mechanism. Each coupler unit shall be provided with a tuning dial mechanism conforming to MIL-STD-454, Requirement 42. Tuning dial accuracy and resettability shall be within ±2.0 percent of indicated frequency. Dial windows shall conform to MIL-E-16400.

3.5.4.2.2 Overload indicators. Each coupler unit shall be equipped with overload indicator lights conforming to MIL-STD-454, Requirement 50.

3.5.4.2.3 Power ON indicator. The antenna coupler group power supply unit shall be equipped with a power ON indicator light conforming to MIL-STD-454, Requirement 50.

3.5.4.2.4 Fuses. Fuses shall be in accordance with MIL-E-16400, except that extra fuses are not required. Fuses shall be mounted on the front panel, in blown fuse indicating fuseholders, which conform to MIL-F-19207. Spare fuseholders are not required.

3.5.4.2.5 Panel lights. Panel lights shall be easily replaceable from the front panel without requiring removal of the unit.

3.5.5 Threaded and thread-locking devices. Threaded and thread-locking devices shall conform to the requirements of MIL-E-16400, except as otherwise specified herein. All screws, nuts, bolts, and other threaded and thread-locking devices shall be of austenitic corrosion-resistant steel, except in applications requiring material capable of withstanding high tensile stresses. Captive-type bolts shall be used on equipment front panels. Cadmium plating is not acceptable on the captive-type bolts used on the equipment front panels.

3.6 Item name and type designation. The item name and type designation shall be Antenna Coupler Group, AN/SRA-56, AN/SRA-57, and AN/SRA-58, and shall include the units specified in a through d:

	AN/SRA-56	AN/SRA-57	AN/SRA-58
a. Electrical equipment cabinet	CY-6422/SRA-56	CY-6423/SRA-57	CY-6424/SRA-58
b. Antenna coupler	CU-1772/SRA-56	CU-1774/SRA-57	CU-1776/SRA-58
c. Power supply	PP-4993/SRA	PP-4993/SRA	PP-4993/SRA
d. Impedance matching network	CU-1773/SRA-56	CU-1775/SRA-57	-------

3.7 Electromagnetic environmental effects (E^3). The antenna coupler shall be self-compatible, operate compatibly in the electromagnetic environment, and, as a minimum, shall conform to the requirements of MIL-STD-461, Class A4. MIL-HDBK-235 shall be used to determine the limits for test RS03. MIL-HDBK-237 may be used to ensure that the electromagnetic compatibility (EMC) considerations are incorporated, and MIL-HDBK-241 may be used for power supply design guidance.

3.7.1 Electromagnetic pulse (EMP). The antenna coupler shall be hardened, or shall be intrinsically hard, against the EMP resulting from an exo-atmospheric nuclear burst to survive the levels specified in 3.7.1.1 and 3.7.1.2.

3.7.1.1 External equipment. Equipment operated external to an intentionally hardened area shall not exhibit any permanent malfunction, degradation of performance, or deviation from specified indications beyond the tolerances and recovery times after being subjected to a test signal having the waveform and amplitude shown in FIGURE 1.

3.7.1.2 Inboard equipment. For equipment operating in an intentionally hardened area, all interface pins and terminals of leads, including grounds and neutrals shall not exhibit any permanent malfunction, degradation of performance, or deviation from specified indications beyond the tolerances and recovery times after being subjected to a test signal having either the waveform and common mode current level shown in FIGURE 2, or the waveform and common mode voltage level shown in FIGURE 2, whichever occurs first at the specific test frequency.

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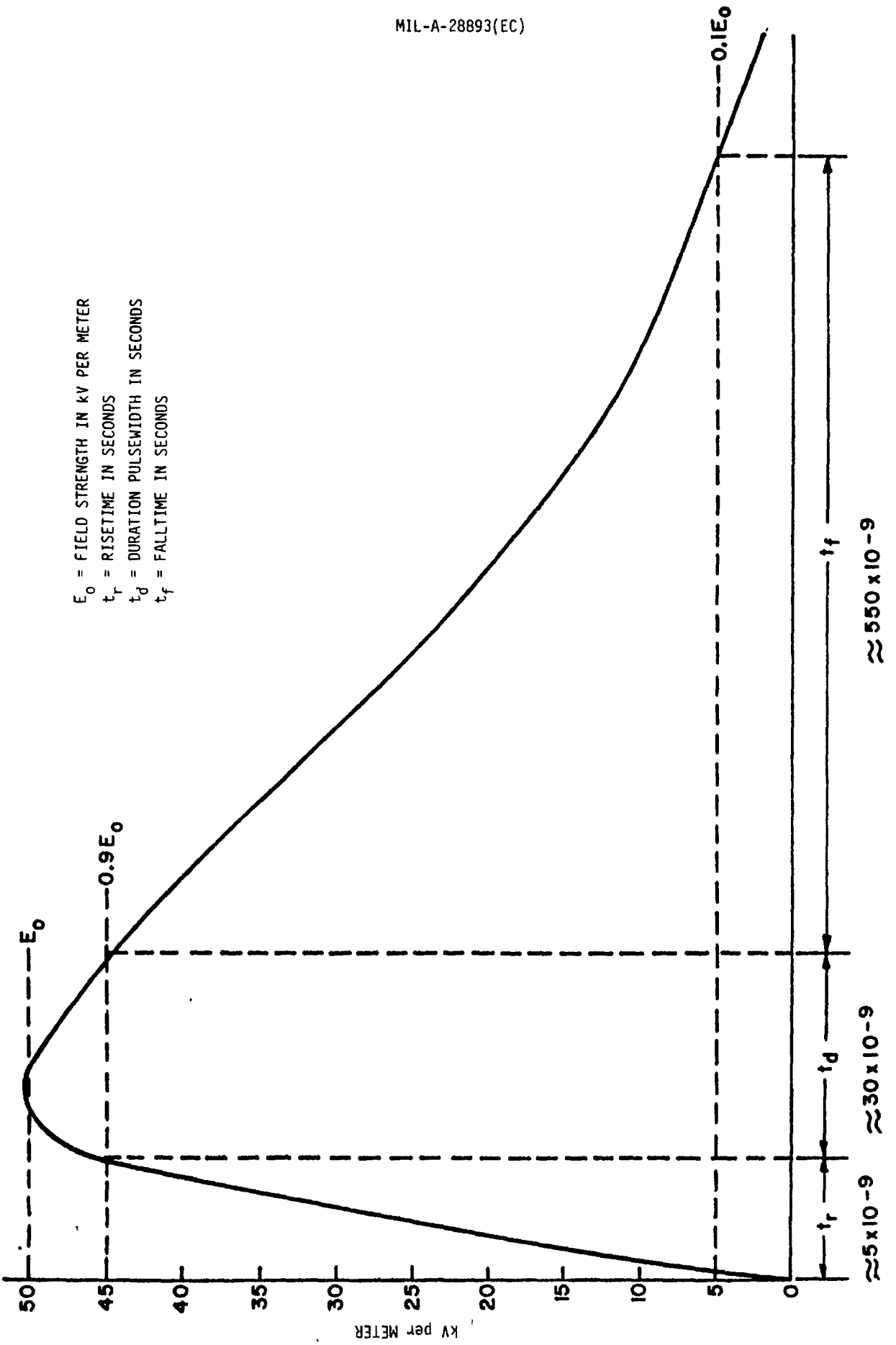
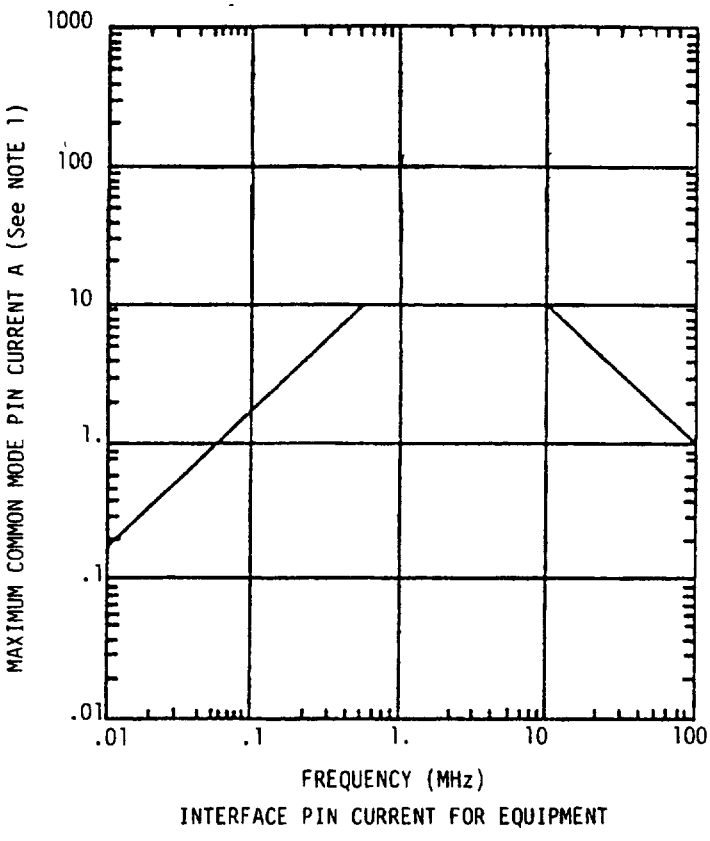
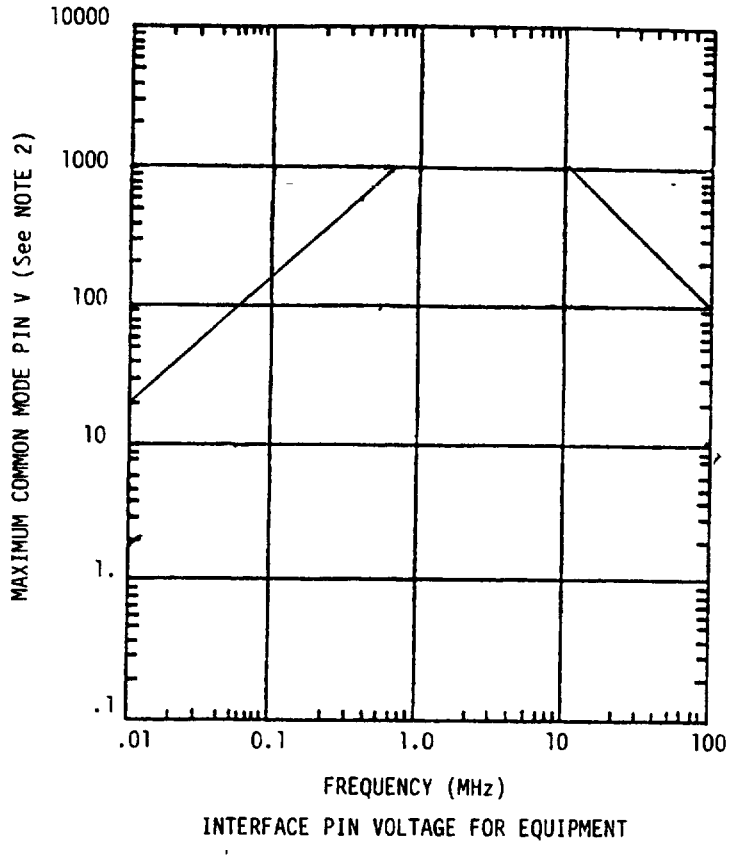


FIGURE 1. Test signal waveform

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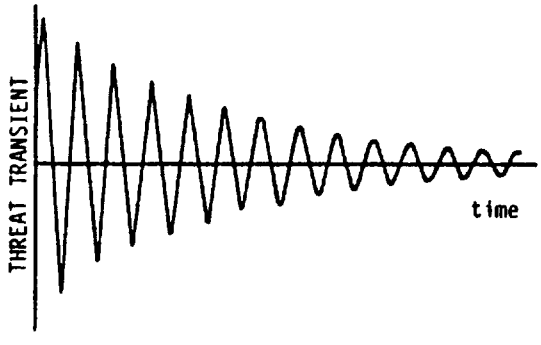
INTERFACE PIN CURRENT FOR EQUIPMENT



INTERFACE PIN VOLTAGE FOR EQUIPMENT

$$x(t) = X(f_o) \text{ Sin } (2\pi f_o t)\ e^{(-\pi f_o t/Q)}$$

$$Q = 16 \pm 4$$



NOTES:

1. Amperes
2. Volts

FIGURE 2. Test signal waveform characteristics.

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3.8 Quantitative reliability requirements. The upper test mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) shall be 3000 hours. The lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 1500 hours.

3.9 Maintainability. The antenna coupler group shall conform to the Maintenance design paragraph of MIL-E-16400. The mean-time-to-repair (MTTR) for the antenna coupler group shall be 1 hour. Test points and test jacks shall be added or incorporated throughout the control unit and coupler to fault locate or troubleshoot using general purpose electronic test equipment. The test points or test jacks shall be used to monitor signals or voltages, or inject signals and may be used in conjunction with existing control unit or coupler meters. The test points or test jacks shall be isolated to the extent necessary to protect equipment circuits from damage when test equipment is connected.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied; b) Government product inspection to measure quality of the product to be offered for acceptance; and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection
    1. Production inspection (Group A) (see 4.4.1)
    2. Production control inspection (Group B) (see 4.4.2)
    3. Environmental inspection (Group C) (see 4.4.3)
c. Inspection of preparation for delivery (see 4.9)

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE IV.

TABLE IV. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Weight	3.2.2.1	4.6	X		X	
Parts and materials	3.1, 3.3.1, 3.3.2	4.6.2	X	X		
Processes	3.1	4.6	X	X		
Dimensions	3.2.2.1	4.6.1	X	X		
Identification and marking	3.1, 3.6	4.6	X	X		
Power handling	3.2.1.1	4.6.3	X	X	X	

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TABLE IV. Examinations and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Frequency range	3.2.1.2	4.6.4	X	X	X	
Passband	3.2.1.3, 3.2.1.3.1, 3.2.1.3.2	4.6.5	X	X	X	
Workmanship	3.1	4.6	X	X		
Isolation						
In-band	3.2.1.4	4.6.6.1	X	X		
Out-of-band	3.2.1.4	4.6.6.2	X	X		
RF overload protection	3.2.1.7	4.6.7	X	X		
Electrical						
Steady-state voltage and frequency	3.4.1.1	4.6	X		X	
Transient voltage	3.1	4.6	X		X	
Transient frequency	3.1	4.6	X		X	
Spike voltage	3.1	4.6	X		X	
Leakage current	3.1	4.6	X		X	
EMI 1/	3.7	4.6.9	X		X	
Temperature	3.2.3.1, 3.2.3.2	4.6	X			X
Humidity	3.1	4.6	X			X
Shock	3.2.3.3	4.6.8	X			X
Vibration	3.1, 3.2.3.4	4.6	X			X
Inclination	3.1	4.6	X			X
Salt fog	3.2.3.5	4.6	X			X
Reliability	3.8	4.7	X			
Maintainability	3.9	4.8	X			
Safety	3.1	4.6	X			
Enclosure	3.1	4.6	X			
Internal wiring and cabling	3.1, 3.4.2	4.6	X			
Bonding and grounding	3.1	4.6	X			

1/ Electromagnetic interference

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.3.2.

4.4.1 Production inspection (Group A). Production inspection shall be performed on every antenna coupler group offered for delivery. Production inspection shall encompass such examinations and tests which will prove the workmanship and reveal any omissions or errors of the production process such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls or adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests shown in Group A of TABLE IV.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis in accordance with MIL-STD-105, using inspection level S-3 with an acceptable quality level of 6.5 percent for each attribute. Production control inspection shall be conducted on inspection lots that passed the production inspection of 4.4.1. Production control inspection shall be performed on the complete antenna coupler group as offered for delivery. Production control inspection shall include the examinations and tests shown in Group B of TABLE IV.

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4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defects or screen out the rejected units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from the new lots and shall retain their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall consist of the tests specified as Group C in TABLE IV.

4.4.3.1 Sampling for environmental inspection. For the first environmental inspection, one antenna coupler group from the first month's production shall be selected. For subsequent environmental inspections, one antenna coupler group per month shall be selected. Tests shall be conducted on lots that have passed the production inspection of 4.4.1. The production control inspection of 4.4.2 shall be performed after the environmental inspection (see 4.4.3).

4.4.3.2 Noncompliance. If a sample antenna coupler group fails environmental inspection, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative (QAR) the results thereof and the details of the corrective action taken to correct units of products which were manufactured with the same conditions, materials, processes, and so forth. If the QAR considers that the corrective action will not enable the antenna coupler group to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. Antenna coupler groups shall not be released for shipment prior to successful resolution of the failure.

4.5 Test conditions. The basic condition specified in 4.5.1 shall apply as pertinent to the performance requirements specified herein, except as modified by a particular test or test procedure contained in this specification.

4.5.1 Ambient temperature. The ambient temperature shall be between 20°Celsius (C) and 30°C.

4.6 Test methods. Examination and tests specified in TABLE IV shall be conducted in accordance with the applicable test methods of MIL-E-16400, except as specified in 4.6.1 through 4.6.9.2.2.3.2.

4.6.1 Dimensions. Antenna coupler group conformance to the dimensions specified in 3.2.2.1 shall be verified with suitable calibrated measuring equipment.

4.6.2 Parts and materials. The antenna coupler group shall be examined to determine conformance to 3.3.1 and 3.3.2.

4.6.3 Power handling capabilities test. The antenna coupler group power handling capability shall be tested to determine conformance to 3.2.1.1

4.6.4 Frequency range test. The antenna coupler group frequency range shall be tested to determine conformance to 3.2.1.2.

4.6.5 Passband test. Each antenna coupler unit shall be tested to determine conformance to 3.2.1.3.1 and 3.2.1.3.2.

4.6.6 Isolation tests. Isolation tests shall be as specified in 4.6.6.1 and 4.6.6.2.

4.6.6.1 In-band isolation test. Each antenna coupler unit shall be tested to determine conformance to 3.2.1.4.

4.6.6.2 Out-of-band isolation test. The antenna coupler group shall be tested to determine conformance to 3.2.1.4.

4.6.7 RF overload tests. Each antenna coupler unit shall be tested to determine conformance to 3.2.1.7.

4.6.8 Shock tests. The antenna coupler group (completely assembled) shall be tested to determine conformance to 3.2.3.3. Upon completion of shock testing, the antenna coupler group shall be examined for visual damage and shall be subjected to the production control inspection (Group B) of 4.4.2 to identify hidden damage resulting from the shock test.

4.6.9 EMI and EMC control tests. The antenna coupler group shall be tested, using method CE03 of MIL-STD-462, to determine conformance to 3.7.

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4.6.9.1 E^3 tests. Conformance to the requirements of 3.7 shall be verified utilizing the test methods of MIL-STD-462.

4.6.9.2 EMP. Tests specified in 4.6.9.2.1 through 4.6.9.2.2.3.2 shall be conducted to determine compliance with 3.7.1.

4.6.9.2.1 External equipment. Conformance to the requirements of 3.7.1.1 shall be verified by an analysis or test of a pulse induced from a simulated exo-atmospheric nuclear explosion whose RF field is of the form and amplitude specified in the following equation:

$$E(t) = E_0(e^{-\alpha t} - e^{-\beta t}) \text{ and}$$

$$H(t) = \frac{E(t)}{Z_0}, \text{ where}$$

E(t) = Electric field in V per meter at time, t
H(t) = Magnetic field in A per meter at time, t
E_0 = 5.25×10^4 V per meter
α = 4.0×10^6 per second
β = 4.76×10^8 per second
Z_0 = 377 ohms
t = Time in seconds

4.6.9.2.2 Inboard equipment. Conformance to the requirements of 3.7.1.2 shall be verified by the test method specified in 4.6.9.2.2.1 through 4.6.9.2.2.3.2.

4.6.9.2.2.1 Applicable test frequencies. Tests shall be performed for the injection frequencies of 0.01 MHz, 0.1 MHz, 0.5 MHz, 1 MHz, 2 MHz, 5 MHz, 10 MHz, 20 MHz, 50 MHz, and 100 MHz. The test shall be performed at the test sample's critical frequencies such as local oscillator, power switching frequencies and harmonics, RF, clock frequencies, bandpass frequencies, and so forth, as specified in the EMI test plan.

4.6.9.2.2.2 Apparatus. Test apparatus shall consist of the equipment specified in a through d:

a. Source injection pulse generator with the capabilities specified in 1 through 4:
1. Damped sinusoidal transient as shown in the applicable limit and as described in the equations in (A) and (B):

A $I_{pin}(t) = 1.05\, I(f)\, e^{-0.2feet\,(ft)} \text{ SIN } (2\pi ft)$

B. $V_{pin}(t) = 1.05\, V(f)\, e^{-0.2ft} \text{ SIN } (2\pi ft)$, where

$I_{pin}(t)$ = Pin to case current, in A at time, t
$V_{pin}(t)$ = Pin to case voltage, in V at time, t
f = Specific test frequency, in MHz
t = Time in seconds
I(f) = Current determined from applicable limit curve of FIGURE 2
V(f) = Voltage determined from applicable limit curve of FIGURE 2

2. Variable amplitude so the injected transient can be increased from a minimum to a maximum specified level (that is, at least 10 percent, 50 percent, and 100 percent of the amplitude shall be tested at each specified test frequency)
3. Frequency range capable of generating the required test frequencies
4. Current output level of at least 10 A into a 100-ohm load

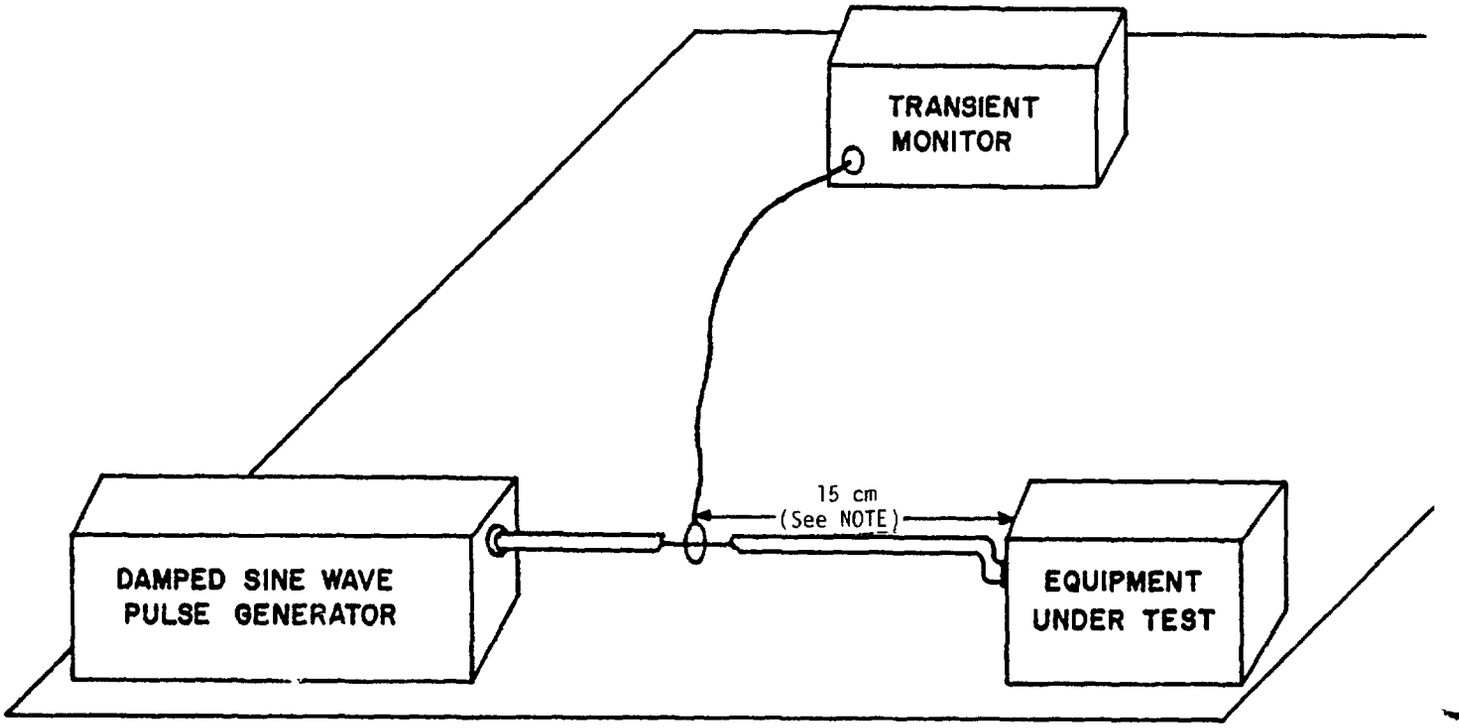
b. Current probe
c. Oscilloscope with a minimum bandwidth capability of 100 MHz
d. Interconnecting test cable

4.6.9.2.2.3 Test setup and procedure. Test setup and procedure shall be as specified in 4.6.9.2.2.3.1 and 4.6.9.2.2.3.2. A functional analysis shall be performed before the certification testing to identify the items specified in a through c:

a. Interface pin connectors
b. Test sample functions which shall be monitored to determine system failure or transient upset
c. Additional test frequencies (that is, interface cable resonant frequency, clock frequency, data rates, and so forth).

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4.6.9.2.2.3.1 Test setup. The test setup shall be as shown in FIGURE 3.



NOTE: Centimeters

FIGURE 3. EMP test setup.

4.6.9.2.2.3.2 Test procedure. The test procedure shall be as specified in a through g:

a. Apply the test signal to each interface pin of the test sample at each test frequency.
b. Monitor the test sample in all modes of operation to determine whether there is a failure or upset.
c. Inject one transient pulse on each interface pin at each test frequency
d. The test sample shall be tested with power off and on.
e. Determine the test frequency and level at which a failure or upset occurs
f. The test sample shall be tested on a ground plane. No insulated materials shall be placed between the test setup and the ground plane.
g. The test cable running between the test sample and the damped sinusoidal generator shall be kept 4 cm to 6 cm above the ground plane.

4.7 Reliability test. A sample of at least one antenna coupler group shall be subjected to reliability testing in accordance with Test Plan XXCI of MIL-STD-781 to demonstrate conformance to 3.8.

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4.7.1 Temperature cycling. Temperature cycling shall be performed between 0°C and +50°C. Time duration at the specified high or low temperature including transition between the two extremes shall be approximately 4 hours.

4.7.2 Vibration cycling. A 4 Hz to 33 Hz to 4 Hz vibration sweep, as specified in the Vibration stress (continuous) paragraph of MIL-STD-781, Appendix B shall be performed throughout the reliability test. The duration of each complete vibration sweep shall be 10 minutes.

4.7.3 Input voltage. Input power shall be nominal during the reliability demonstration test.

4.7.4 Operation duty cycle. The equipment shall be nonoperating during the first 30-minute period each 4-hour test period. The equipment shall be on and operating during the remainder of the test. Maximum rated RF power shall be applied to the input.

4.7.5 Performance measurements. At least daily, insertion loss and isolation measurements shall be made to verify proper equipment operation within the constraints specified in 3.2.1.3.1 and 3.2.1.4, respectively.

4.7.6 Failure. The procuring activity shall be notified of any failure during reliability testing. The procuring activity shall classify failures in accordance with the Failure categories paragraph of MIL-STD-781.

4.7.7 Rejected equipment. If equipment is rejected as a result of reliability testing, the contractor shall immediately notify the procuring activity of the cause(s) for rejection and propose a plan for correction of the deficiencies. When such corrections have been approved by the procuring activity and incorporated in the test unit, the reliability test shall be continued as specified herein.

4.8 Maintainability demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be performed by qualified technicians and shall be used to verify the antenna coupler group mean-time-to-repair (MTTR) requirement is fulfilled.

4.8.1 Demonstration conditions. The maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual printed circuit boards) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. One hundred candidate corrective maintenance faults shall be determined in accordance with Appendix A of MIL-STD-471. The information specified in a through c shall be made available for each candidate task:

a. Designation of specific faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part or simulation thereof)

The procuring activity or its authorized representative, shall use the candidate faults as a guide to select a sample of 50 faults for the demonstration.

4.8.2 Accept or reject criteria. Accept or reject criteria shall be as specified in 4.8.2.1 through 4.8.2.4.

4.8.2.1 Corrective maintenance. The accept or reject criteria for the demonstration of the corrective maintenance times for the demonstration shall be as specified in TABLE V.

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TABLE V. Acceptance criteria.

	MTTR	M_{maxct} 1/
Acceptance level	20	0
Sample size	50	50

1/ Maximum-corrective-maintenance-time

Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value of each specified index (MTTR or M_{maxct}) is less than or equal to that shown in TABLE V corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s) and determined as greater than or less than each index. The actual value shall also be determined. An accept decision can only be made when an accept decision is made for both MTTR and M_{maxct}. Accept or reject criteria for preventive maintenance shall be the capability to perform each preventive maintenance task with no degradation of system performance. If degradation of system operations exists, the preventive maintenance test shall have resulted in a reject decision. More than one erroneous failure indication shall constitute failure of the demonstration.

4.8.2.2 Preventive maintenance. The demonstration shall qualitatively demonstrate that on-line preventive maintenance for the antenna coupler group shall not interfere with on-line operation.

4.8.2.3 Acceptance. The maintainability demonstration shall demonstrate that the antenna coupler group conforms to the accept criteria prior to delivery.

4.8.2.4 Rejection. Failure to conform to any of the maintainability requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the contracting officer shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies
b. Upon approval of an approach to the redesign or corrections, the contractor shall implement the approach and repeat the demonstration until an accept decision is reached.

4.8.3 Demonstration conditions. Demonstration conditions shall be as specified in 4.8.3.1 through 4.8.3.3.

4.8.3.1 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the antenna coupler group will be installed.

4.8.3.2 Technicians. The procuring activity reserves the right to provide and select technicians to perform the maintainability demonstration.

4.8.3.3 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation delivered with the antenna coupler group.

4.9 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

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6. NOTES

6.1 Intended use. The antenna coupler group covered by this specification is intended for use aboard surface ships for general communications purposes.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. Levels of preservation, packaging, packing, and marking (see 5.1)

6.3 First article. When a first article is required it should be tested and approved under the appropriate provisions of 52.209-3 of the Federal Acquisition Regulations. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examinations, tests, and approval of the first article.

6.4. Supersession data. This specification includes the requirements of SHIPS-M-5432, dated 31 January 1969.

Preparing activity:
NAVY - EC

(Project No. 5985-N522)

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5. PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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